

**AI Programming Foundations Capstone Project: Analyzing Airbnb Listings in New York
City**

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Abstract

Replace this text with your abstract. Summarize the research problem, method, main findings, and conclusion. Add keywords below when required by your institution.

Keywords: keyword one, keyword two, keyword three

Overview

This project implements a reproducible Python workflow for cleaning, exploring, and visualizing the New York City Airbnb Open Data dataset (Gomonov, 2019). The dataset is available from Kaggle at <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>. The analysis examines how listing prices and annual availability vary across New York City neighbourhood groups and room types. The complete workflow is contained in the accompanying `data_workflow.ipynb` notebook.

Dataset Description

The New York City Airbnb Open Data dataset contains listing-level information for Airbnb properties in New York City during 2019. The original CSV file contains 48,895 rows and 16 columns, with each row representing one listing.

The analysis focused primarily on `price`, `neighbourhood_group`, `room_type`, `availability_365`, `minimum_nights`, `number_of_reviews`, `reviews_per_month`, and `calculated_host_listings_count`. Latitude, longitude, listing identifiers, listing names, and host information were retained but were not central to the main comparisons.

The dataset includes the five neighbourhood groups Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Its room-type categories are entire home or apartment, private room, and shared room.

Workflow Description

The project followed a sequential workflow consisting of data ingestion, inspection, cleaning, exploratory analysis, visualization, and interpretation. This structure was chosen to make the analytical process transparent and repeatable. Danchev describes reproducibility as a principle that should be incorporated throughout the data-science lifecycle rather than treated only as a final documentation activity (Danchev, 2022).

Ingestion

The CSV file was stored within the repository and loaded with `pandas.read_csv()`. The notebook then inspected the dimensions, column names, data types, missing values, duplicated rows, categorical values, and numerical ranges before applying any transformations.

Cleaning

Two reusable cleaning functions were created. The first converted `last_review` to a datetime type and handled missing review-rate values. The second removed records that violated explicit validity rules, including non-positive prices.

Exploratory Analysis

A reusable exploratory-analysis function generated numerical summaries, neighbourhood-level summaries, room-type summaries, combined neighbourhood and room-type comparisons, and a Pearson correlation matrix. Additional analyses examined zero availability and compared mean and median prices before and after limiting prices to the 99th percentile.

Visualizations

Four visualizations were created:

1. median price by neighbourhood group and room type;
2. price distributions by room type;
3. median annual availability by neighbourhood group and room type;
4. correlations between numerical variables.

Summary

The final notebook section interpreted the calculated results, documented the main limitations and assumptions, and identified questions that could not be resolved from the available variables.

Key Decisions and Assumptions

Missing Review Information

The original dataset contained 10,052 missing values in both `last_review` and `reviews_per_month`. Inspection showed that these missing values were associated with listings that had zero recorded reviews.

Missing `reviews_per_month` values were therefore replaced with zero only where `number_of_reviews` was also zero. This preserved unreviewed listings while representing their observed monthly review activity numerically. Missing `last_review` values were retained because a listing with no reviews has no meaningful last-review date.

This variable-specific approach was preferable to deleting every incomplete record. Missing-data treatment can affect representativeness and introduce bias, so the meaning and pattern of missingness should be considered before selecting a treatment (Hyun, 2013).

The small numbers of missing listing names and host names were retained because those descriptive fields were not needed for the principal price and availability analyses.

Invalid and Extreme Values

The cleaning process removed 11 records whose prices were zero. A non-positive nightly price was treated as invalid for the planned price analysis. This reduced the dataset from 48,895 to 48,884 listings.

Large positive prices and long minimum-stay requirements were not automatically removed. Although these observations are unusual, the dataset provides insufficient evidence to classify them as errors. Removing them without an explicit validity rule could exclude legitimate listings and alter the observed distribution.

The price distribution was strongly right-skewed. The full cleaned dataset had a mean price of \$152.76 and a median price of \$106.00. For listings priced at or below the 99th percentile of \$799, the mean decreased to \$137.58, whereas the median changed only to \$105.00. Median prices were consequently used for the main grouped comparisons because they were less sensitive to extreme values.

Prices above the 99th percentile were excluded only from Figure 2 in the accompanying notebook to improve its readability. They remained in the cleaned dataset and in the other analyses.

Choice of Exploratory Comparisons

Neighbourhood group and room type were selected as the central categorical variables because they allowed direct comparison of location and accommodation category. Price and availability were grouped by both variables so that their combined patterns would not be hidden by separate aggregate summaries.

The correlation matrix was included to examine linear relationships among the numerical variables. Correlation coefficients were interpreted as descriptive associations and not as evidence of causation.

References

- Danchev, V. (2022). Reproducible data science with python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>
- Gomonov, D. (2019). *New york city airbnb open data* [Accessed 2026-08-04]. Kaggle. <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>
- Hyun, K. (2013). The prevention and handling of the missing data. *Korean J Anesthesiol*, 64(5), 402–406. <https://doi.org/10.4097/kjae.2013.64.5.402>

Supplementary Material

Place questionnaires, extended tables, source code excerpts, or other supporting material here when appropriate.